Estimation

1 State whether the following sums are approximately equal to 10, 100, or 1000:

a $23 + 42 + 34$ **b** $93 + 342 + 567$ **c** $364 + 292 + 311$ **d** $6 + 3 + 3$

2 State whether the following sums are approximately equal to 10:

a $4 + 2 + 3$ **b** $9 + 8 - 14$ **c** $2 + 8 + 19$ **d** $9 + 9 - 7$

3 State whether the following sums are approximately equal to 100:

a $28 + 32 + 43$ **b** $99 + 88 - 78$ **c** $131 + 62 + 11$ **d** $45 + 12 - 7$

4 State whether the following sums are approximately equal to 1000:

a $452 + 213 + 334$ **b** $99 + 188 - 68$

c $948 + 827 + 18$ **d** $946 + 28 + 14 + 19$

5 State whether the following statements are reasonable:

a $59 - 27$ is approximately equal to 30.

b $68 - 27$ is approximately equal to 60.

c $2883 - 2831$ is approximately equal to $80 - 30$.

d $6871 - 5832$ is approximately equal to $70 - 30$.

6 State whether the following statements are accurate:

a 4×89 will be less than 320. **b** 3×85 will be more than 240.

7 Consider the product 37×94 and determine which of the following statements are correct:

a The product is less than 4000. **b** The product is more than 2700.

c The product is more than 4000. **d** The product is less than 2700.

8 State whether the following products are approximately equal to 1200:

a 33×38 **b** 31×18 **c** 203×6 **d** 68×69

9 State whether $821 \div 400$ is approximately equal to 200.

10 State whether $4 \div 1221$ is approximately equal to 300.

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11 Approximate the following calculations to the nearest 100:

a 45×93 **b** 178×5 **c** $1621 \div 4$ **d** $5513 \div 11$

12 Fred says "Dividing by 6 is the same as dividing by 3 and then dividing by 3 again". Is Fred correct? Explain your answer.

13 Emma says "Multiplying by 15 is the same as multiplying by 3 and then multiplying by 5 ". Is Emma correct?

Applications

14 Michael collects padlocks. He knows that he has exactly 195 of them. Michael has to move house this week, and must buy storage boxes to transport them.
Should Michael buy 7 boxes that hold 30 padlocks each or 5 boxes that hold 30 padlocks each?

15 Ryan and Michael went out one evening. Michael said:
"I have $\$17.40$, which is about $\$20$. This restaurant charges $\$9.75$ for a pizza, which is about $\$10$. That means I have enough money to pay for both of us to get a pizza."

a Is Michael correct? Explain your answer.

b Ryan says, "I can put $\$5$ towards our meal, then we can both afford a pizza." Is Ryan correct? Explain your answer.

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